

String Packages

In Go, you rarely write custom loops to search or modify text. Instead, Go provides three highly optimized packages (unicode, strings, and strconv) to handle almost every text-processing scenario imaginable.

1. The Unicode Package (Character-Level)

This package is used when you need to inspect or modify **individual runes** (single characters) inside a string. This is heavily used when parsing raw user input character-by-character.

- **Evaluation Functions (Returns True/False):**

- **IsDigit(rune):** Is it a number (0-9)?
- **IsSpace(rune):** Is it a space, tab, or newline?
- **IsLetter(rune):** Is it an alphabetical letter?
- **IsPunct(rune):** Is it punctuation?

- **Conversion Functions:**

- **ToUpper(rune) / ToLower(rune):** Converts a single lowercase rune to uppercase, or vice versa.

2. The strings Package (String-Level)

This package is used when you want to evaluate or manipulate the **entire string** at once.

- **Search Functions:**

- **Compare(a, b):** Compares two strings alphanumerically. Returns 0 if equal, -1 if a comes first, and 1 if b comes first.
- **Contains(s, substring):** Returns true if the substring exists anywhere inside the main string.
- **HasPrefix(s, prefix):** Returns true if the string starts with the specific prefix.
- **Index(s, substring):** Searches for the substring and returns the exact index (position) where it first appears.

- **Manipulation Functions:**

- **Replace(s, old, new):** Swaps occurrences of an 'old' word with a 'new' word.
- **ToUpper(s) / ToLower(s):** Converts the entire string's case.
- **TrimSpace(s):** Extremely useful for cleaning up text. It strips away any invisible spaces, tabs, or newlines at the very beginning and very end of the string.

Remember: Strings are read-only! These functions do not change your original string; they return a brand-new, modified string.

3. The strconv Package (Type Conversions)

Because Go is strictly typed, a string like "123" is just text. You cannot do math on it. The **strconv** (String Conversion) package is how you translate text into actual usable data types, and vice versa.

- **Integers:**
 - **Atoi (ASCII to Integer):** Converts a **string** to an **int** (e.g., "123" becomes the number 123).
 - **Itoa (Integer to ASCII):** Converts an **int** back into a **string**.
- **Floating Point:**
 - **ParseFloat:** Converts a string into a **float64** (e.g., "123.45" becomes a usable decimal number).
 - **FormatFloat:** Converts a float64 into a string representation.